Module 9 · Lecture 1

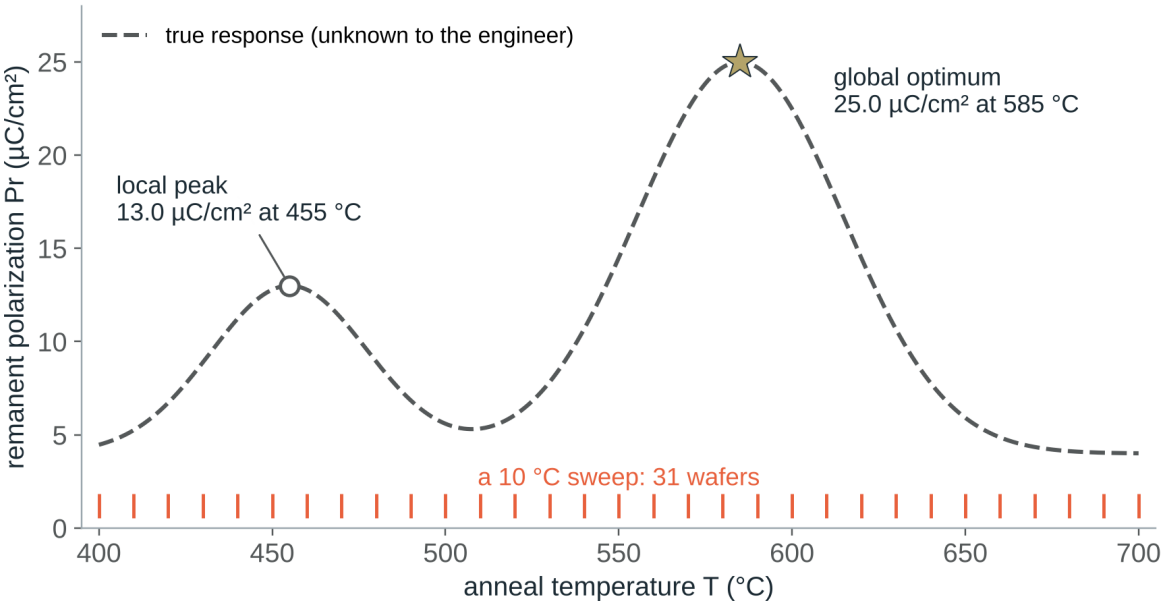
Bayesian Optimization: The Math, Step by Step

From a Gaussian random variable to expected improvement, with a seven-measurement worked example

Asif Khan Associate Professor · School of ECE and School of MSE
Georgia Institute of Technology · ECE 8803-AI4 · akhan40@gatech.edu

Georgia Tech

The problem: the best anneal temperature



A ferroelectric film

Knob: anneal temperature T, 400–700 °C.

Response: remanent polarization Pr, to be maximized.

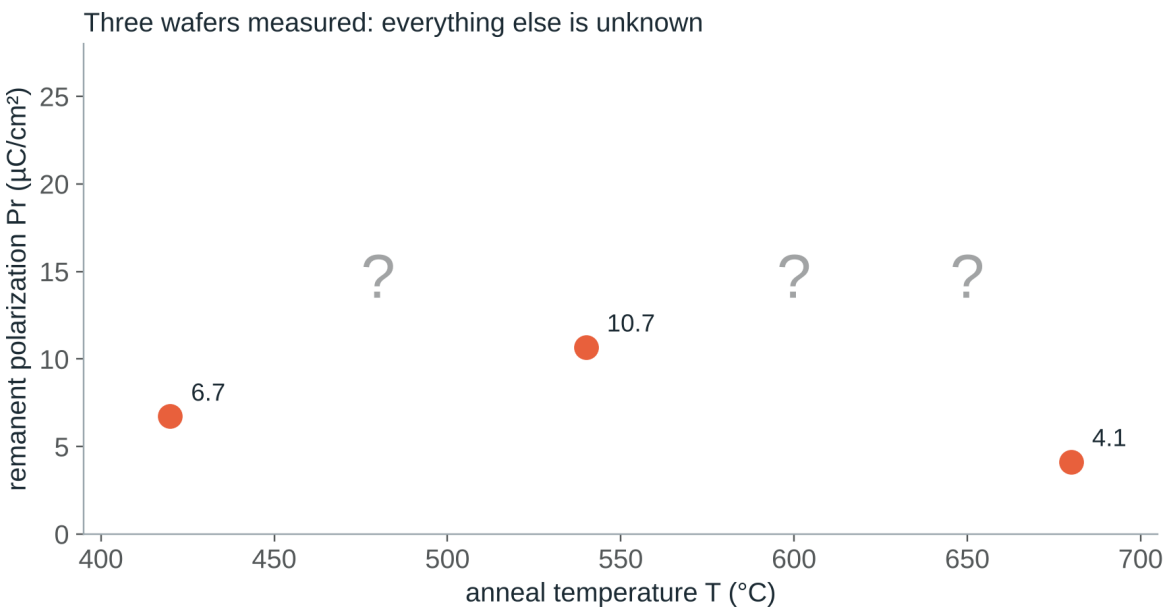
Cost: one wafer per temperature. A 10 °C sweep needs 31 wafers.

Trap: a local peak of 13.0 µC/cm² at 455 °C; the optimum is 25.0 µC/cm² at 585 °C.

Takeaway The curve is unknown to the engineer. Each point on it costs a wafer.

Georgia Tech

Three wafers in: what do we know?



1 **What is Pr at an untried T, and how sure are we?**

Prediction with an error bar

2 **Which temperature should wafer four use?**

An acquisition rule

3 **When do we stop?**

A stopping rule

Takeaway This lecture derives the answer to each question from a Gaussian random variable up.

Georgia Tech

What you'll take away

1. **Gaussian variables and correlation:** two numbers fix a distribution; correlation links one setting to the next.
2. **The kernel:** the covariance of f(x) and f(x′), set by one correlation length ℓ, like the autocovariance of filtered noise.
3. **Prediction:** after each measurement, a mean and a standard deviation at every untried setting, from least squares.
4. **Expected improvement:** the average gain over the best result, in closed form, and how it balances exploration and exploitation.
5. **Worked examples:** seven measurements on a test function with a stopping rule, then wafer thickness from metrology sites.

Georgia Tech

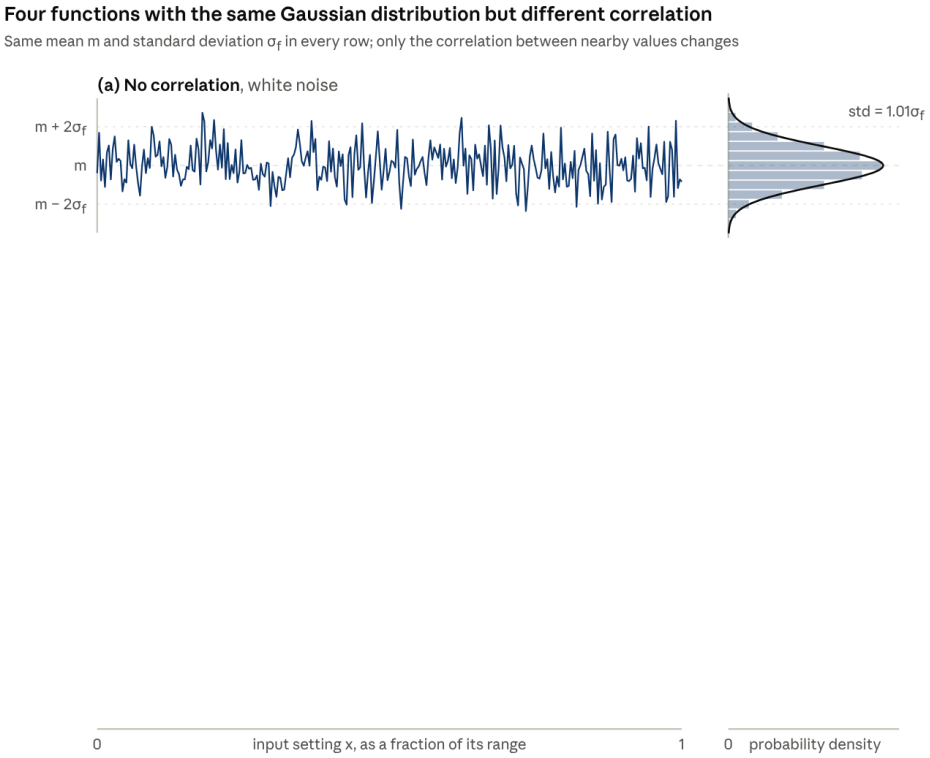
A

BACKGROUND

Gaussian variables and correlation

Georgia Tech

Four functions with the same distribution



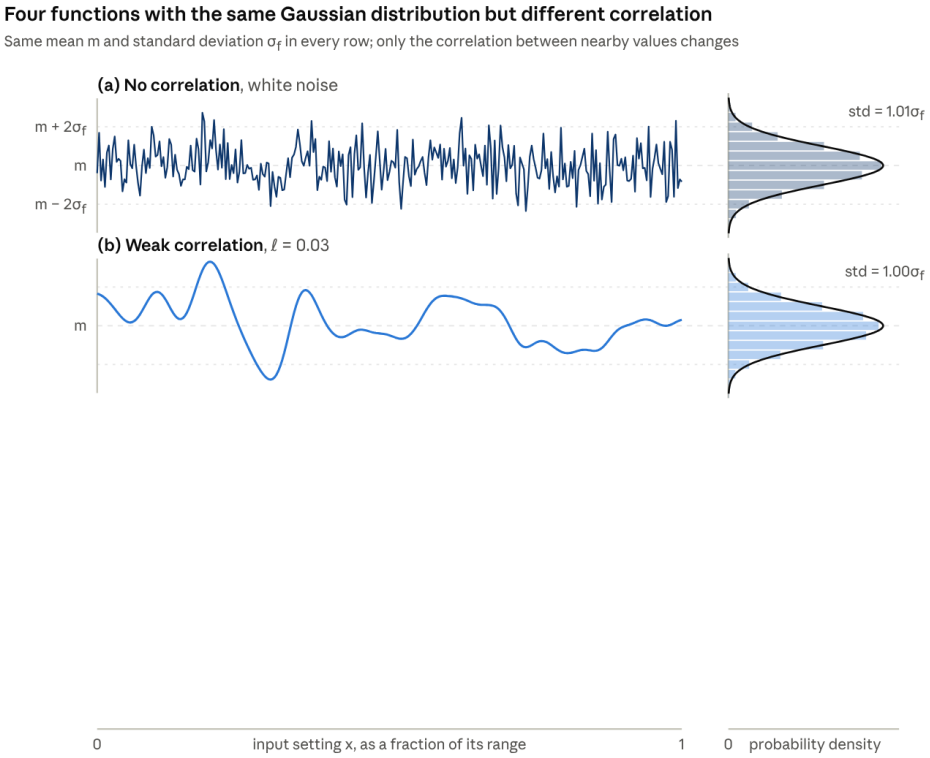
Reading the figure

Left: f(x) across the input range.

Right: histogram of f over a record 1000 times longer, with the Gaussian density on top.

(a) White noise: each value independent of its neighbors; std $1.01\sigma_f$.

Four functions with the same distribution



Reading the figure

Left: f(x) across the input range.

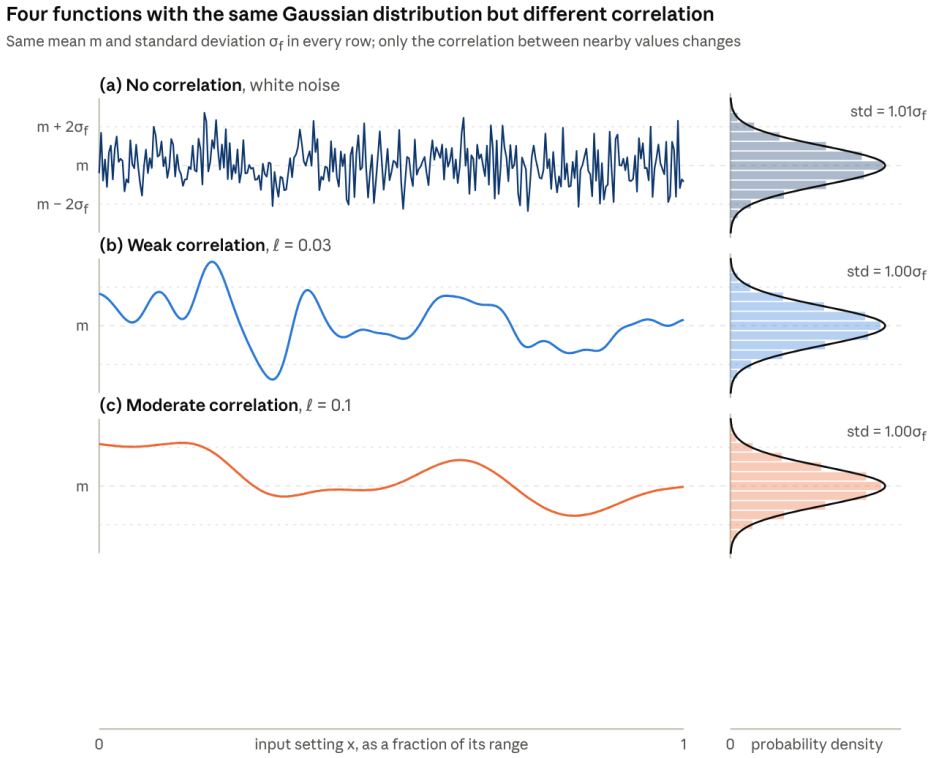
Right: histogram of f over a record 1000 times longer, with the Gaussian density on top.

(a) White noise: each value independent of its neighbors; std $1.01\sigma_f$.

(b) the same noise through a low-pass filter, $\ell = 0.03$; std $1.00\sigma_f$.

Georgia Tech

Four functions with the same distribution



Reading the figure

Left: f(x) across the input range.

Right: histogram of f over a record 1000 times longer, with the Gaussian density on top.

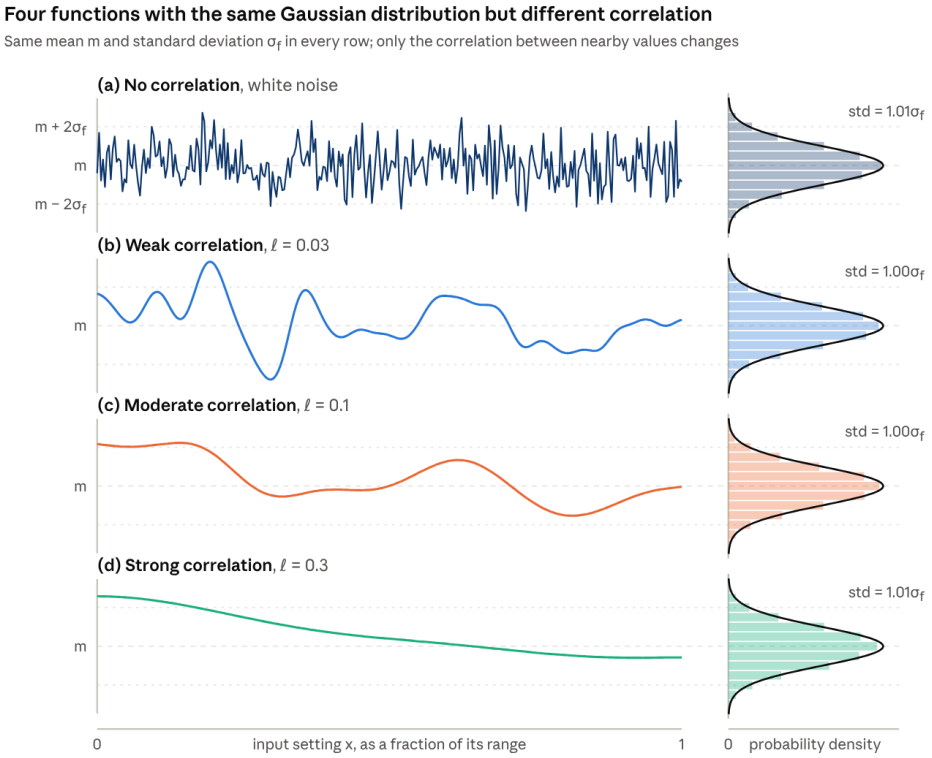
(a) White noise: each value independent of its neighbors; std $1.01\sigma_f$.

(b) the same noise through a low-pass filter, $\ell = 0.03$; std $1.00\sigma_f$.

(c) a longer time constant, $\ell = 0.1$; std $1.00\sigma_f$.

Georgia Tech

Four functions with the same distribution



Reading the figure

Left: f(x) across the input range.

Right: histogram of f over a record 1000 times longer, with the Gaussian density on top.

(a) White noise: each value independent of its neighbors; std $1.01\sigma_f$.

(b) the same noise through a low-pass filter, $\ell = 0.03$; std $1.00\sigma_f$.

(c) a longer time constant, $\ell = 0.1$; std $1.00\sigma_f$.

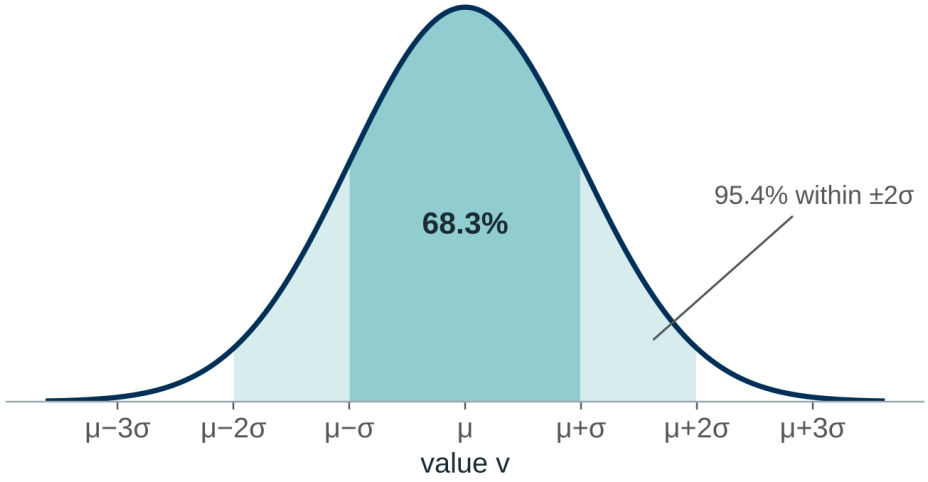
(d) longer still, $\ell = 0.3$; std $1.01\sigma_f$.

Takeaway A histogram at one setting cannot tell the four rows apart. The link between nearby settings can.

Georgia Tech

The Gaussian random variable

$$p(v) = \frac{1}{\sqrt{2\pi}\,\sigma}\,\exp\left(-\frac{(v-\mu)^2}{2\sigma^2}\right)$$



Reading the density

μ sets the center. **σ**, the RMS deviation from the mean, sets the width; σ² is the variance. Shorthand: v ~ N(μ, σ²).

Probability within ±σ: 68.3%; ±2σ: 95.4%; ±3σ: 99.7%.

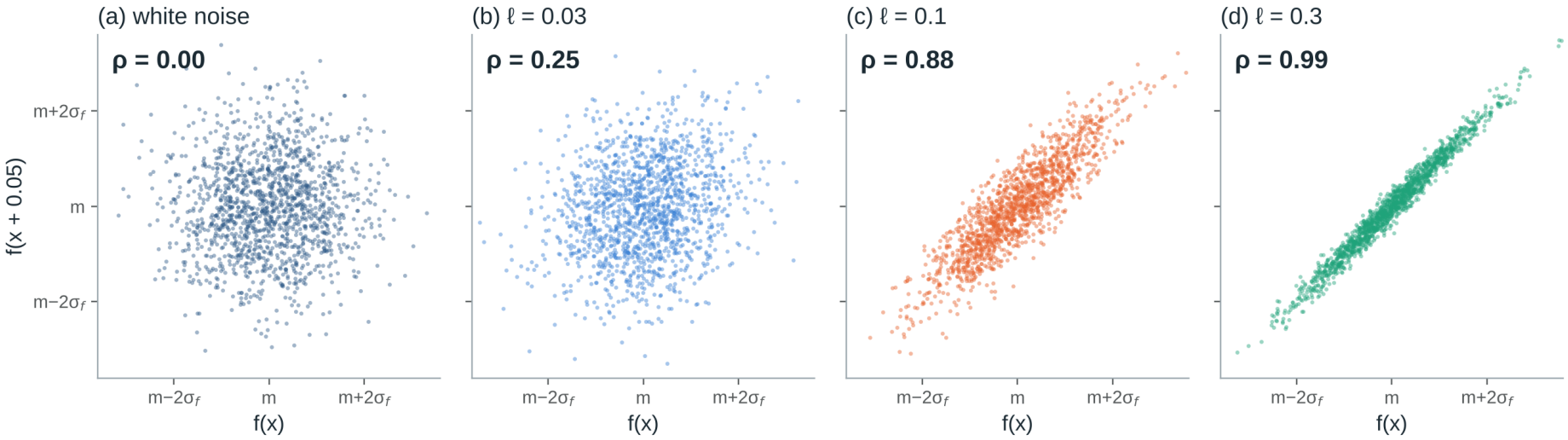
Example: the thermal noise voltage across a resistor. It sums many independent electron motions, so it is Gaussian (central limit theorem).

Takeaway Two numbers, μ and σ, fix the whole distribution. In every row of the last figure, μ = m and σ = σ_f.

Georgia Tech

Covariance and correlation

$$c = \mathrm{Cov}(A, B) = \mathbb{E}\big[(A - \mu_A)(B - \mu_B)\big] \qquad \rho = \frac{c}{\sigma_A \, \sigma_B}, \qquad -1 \le \rho \le 1$$

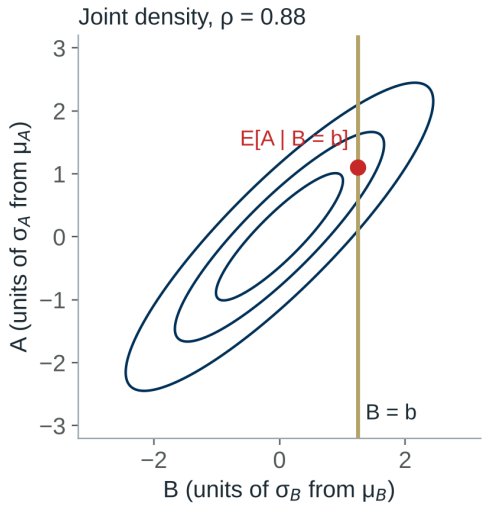


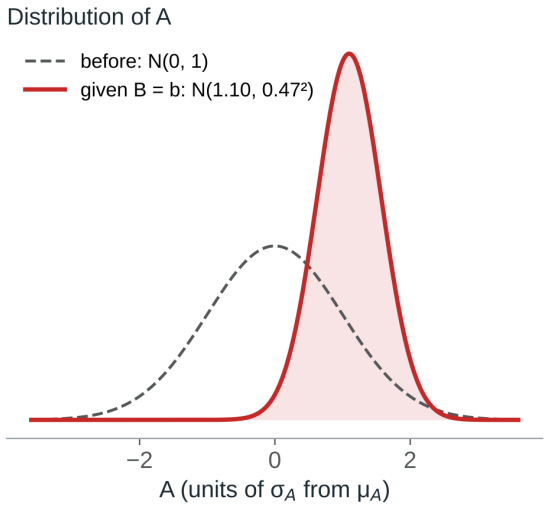
Takeaway For f(x) and f(x + 0.05), ρ = 0.00, 0.25, 0.88 and 0.99 in rows (a)–(d). The histograms hid all of it.

Georgia Tech

An observation of B updates A

$$\mathbb{E}[A \mid B = b] = \mu_A + \frac{c}{\sigma_B^2}(b - \mu_B), \qquad \mathrm{Var}[A \mid B = b] = \sigma_A^2 (1 - \rho^2)$$





Example: ρ = 0.88

Observation: $b = \mu_B + 1.25\sigma_B$.

Mean of A: $\mu_A \to \mu_A + 1.10\sigma_A$.

Spread of A: $\sigma_A \to 0.47\sigma_A$.

ρ = 1: B fixes A exactly. **ρ = 0:** B says nothing about A.

Takeaway Applied to f(x′) and a measured f(x), this formula is the prediction step of Bayesian optimization.

Georgia Tech

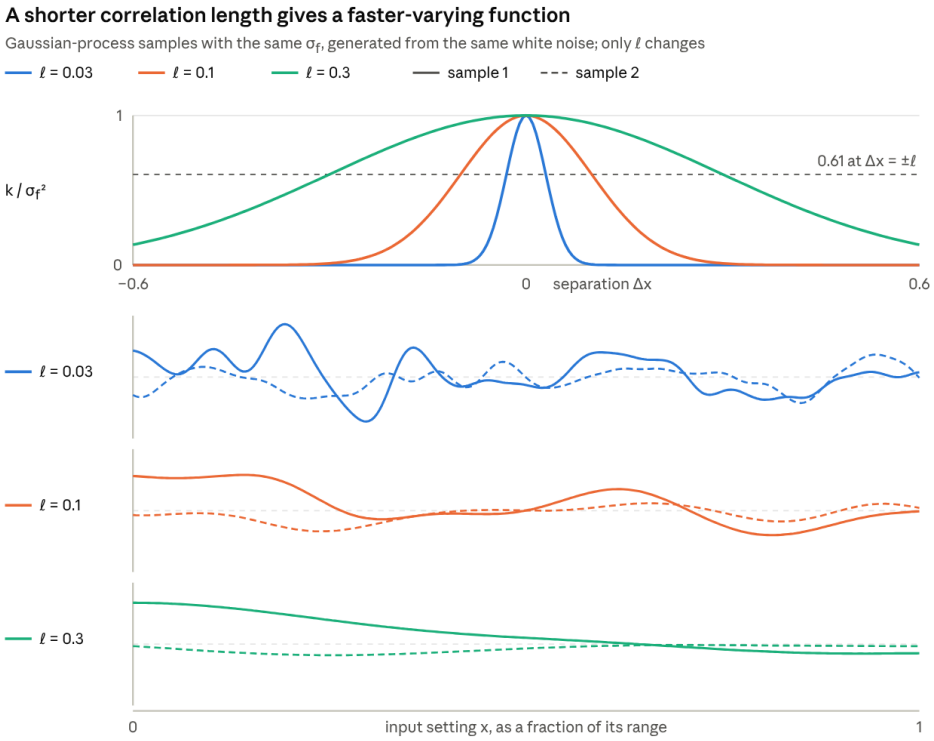
B

THE KERNEL

A Gaussian process and its correlation length

Georgia Tech

The kernel: covariance as a function of separation



$$k(x, x') = \mathrm{Cov}\left[f(x),\ f(x')\right]$$

$$k(\Delta x) = \sigma_f^2 \exp\left(-\frac{\Delta x^2}{2\ell^2}\right)$$

Gaussian process

f(x) is Gaussian at every x; values at any set of settings are jointly Gaussian.

Correlation $\rho(\Delta x) = k/\sigma_f^2$: 0.61 at $\Delta x = \ell$, 0.14 at 2ℓ, 0.011 at 3ℓ.

EE view: k is the autocovariance of a noise voltage; the process is white noise through a Gaussian low-pass filter of width $\ell/\sqrt{2}$.

Takeaway ℓ is the correlation length: values closer than about ℓ move together; beyond about 3ℓ they are independent.

Georgia Tech

Notation

Symbol	Meaning
x, x'	Input settings: x measured, x' a candidate not yet measured
$f(x)$	True response at x; unknown until measured, so treated as a random variable
m, σ_f	Prior mean and prior standard deviation of f
ℓ	Correlation length of the kernel
$k(x', x)$	Kernel: the covariance of $f(x')$ and $f(x)$
$m'(x')$	Posterior mean: $m'(x') = E[f(x') \mid \text{data}]$, the best estimate after measurement
$\sigma'(x')$	Posterior standard deviation: $\sigma'(x')^2 = \text{Var}[f(x') \mid \text{data}]$
y^+	Largest value measured so far (the incumbent)
$\alpha(x')$	Acquisition function: the expected gain from one measurement at x'
φ, Φ	Standard normal density and its cumulative area

Georgia Tech

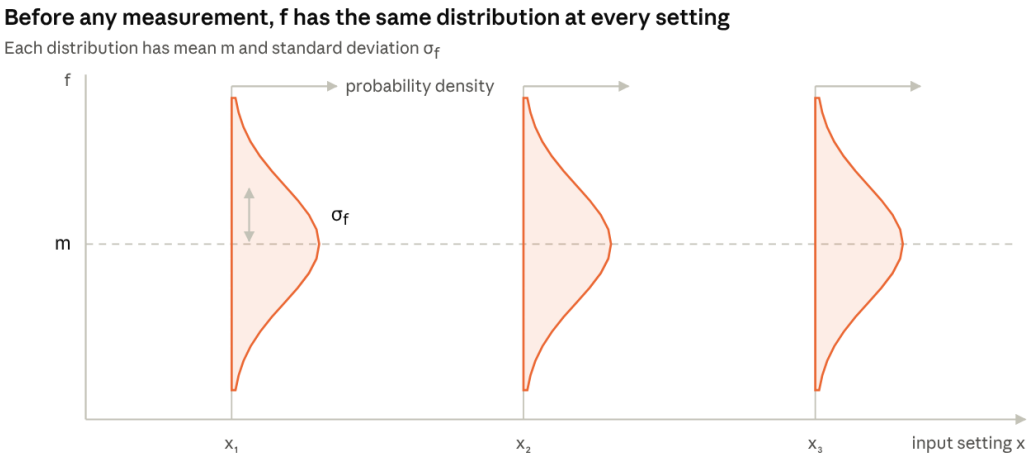
C

PREDICTION

What one measurement tells you

Georgia Tech

Steps 1–2: the objective and the prior



Step 1 · Objective

$$x^\star = \arg\max_x \; f(x)$$

Step 2 · Prior

Before any data, f(x) at every setting is Gaussian with mean m and standard deviation σ_f.

$$p\big(f(x)\big) = \frac{1}{\sqrt{2\pi}\,\sigma_f} \exp\left(-\frac{\big(f(x)-m\big)^2}{2\sigma_f^2}\right)$$

Takeaway Before the first measurement, every setting has the same distribution, $N(m, \sigma_f^2)$.

Georgia Tech

Steps 3–4: the best estimate after one measurement

1 **Write the estimate as a straight line in f(x), with one unknown slope w**

$$m'(x') = \mathbb{E}\left[f(x') \mid f(x)\right] = m + w\left(f(x) - m\right)$$

2 **Its mean-squared error, using Var[f] = σ_f^2 and Cov[f(x′), f(x)] = k(x′, x)**

$$\mathbb{E}\left[\left(f(x') - m'(x')\right)^2\right] = \sigma_f^2 - 2w\,k(x', x) + w^2\sigma_f^2$$

3 **Set the derivative with respect to w to zero**

$$-2\,k(x', x) + 2w\,\sigma_f^2 = 0 \quad \Rightarrow \quad w = \frac{k(x', x)}{\sigma_f^2}$$

Why this is optimal

With this w, the remaining error is uncorrelated with f(x).

For Gaussian variables, uncorrelated means independent.

So no estimator, linear or not, predicts f(x′) better from f(x).

Takeaway The estimate moves from m toward f(x) by the fraction w = k(x′, x)/σ_f^2.

Georgia Tech

The posterior mean and standard deviation

$$m'(x') = m + \frac{k(x', x)}{\sigma_f^2}\big(f(x) - m\big), \qquad \sigma'(x')^2 = \sigma_f^2 - \frac{k(x', x)^2}{\sigma_f^2}$$

With the squared-exponential kernel, σ_f^2 cancels from the mean:

$$m'(x') = m + e^{-(x'-x)^2/2\ell^2}\,\big(f(x) - m\big)$$

$$\sigma'(x')^2 = \sigma_f^2\left(1 - e^{-(x'-x)^2/\ell^2}\right)$$

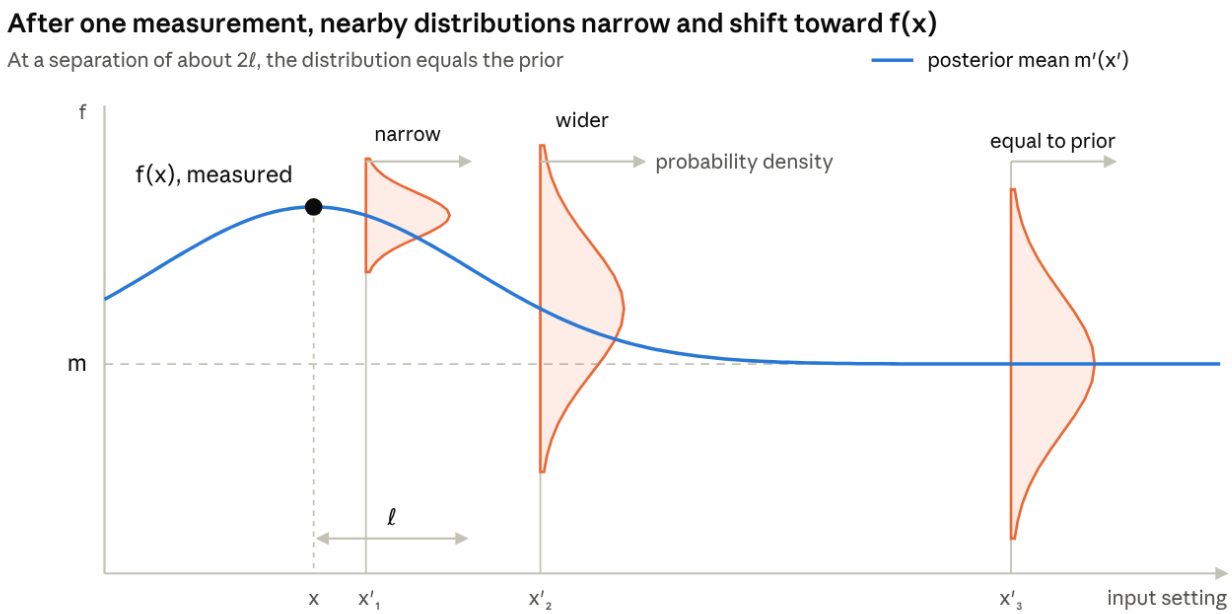
Separation	m′(x′)	σ′(x′)
$\lvert x' - x\rvert \ll \ell$	≈ f(x)	≈ 0
$\lvert x' - x\rvert = \ell$	m + 0.61 (f(x) − m)	0.80 σ_f
$\lvert x' - x\rvert = 2\ell$	m + 0.14 (f(x) − m)	0.99 σ_f
$\lvert x' - x\rvert \gg \ell$	≈ m	≈ σ_f

Each untried setting x′ now has its own Gaussian: mean m′(x′), standard deviation σ′(x′).

Takeaway Within a fraction of ℓ of the measurement you know f; beyond about 2ℓ you are back to the prior.

Georgia Tech

After one measurement: a distribution at every setting



Reading the figure

Blue curve: the posterior mean m′(x′).

Each bell: the distribution of f(x′) at one candidate, of width σ′(x′).

Near x: narrow, centered close to f(x).

About 2ℓ away: the prior distribution again.

Takeaway The prediction is a full distribution at every candidate, which is what the next step needs.

Georgia Tech

D

ACQUISITION

Expected improvement: where to measure next

Georgia Tech

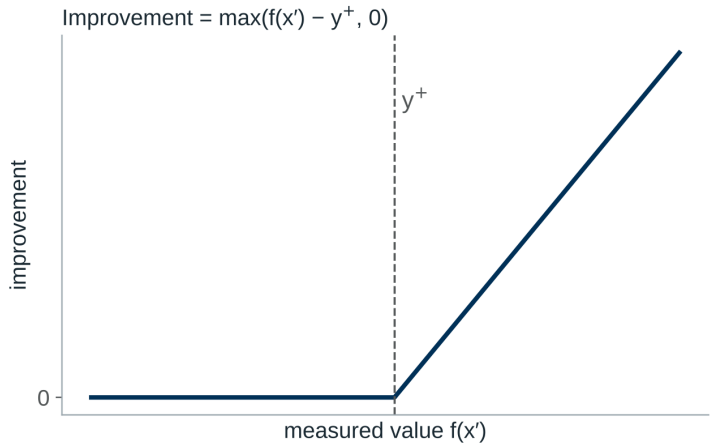
Steps 5–7: improvement and its average

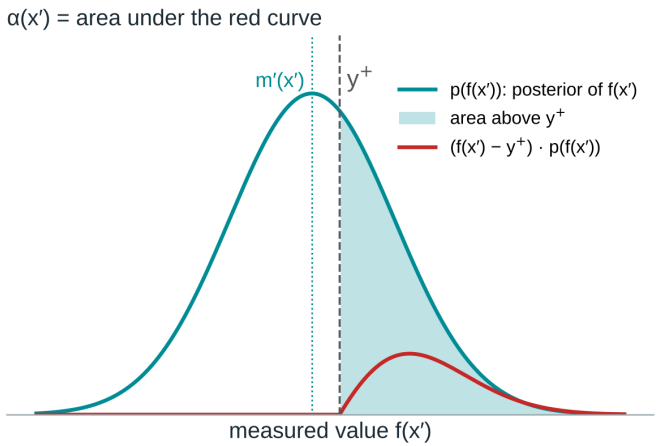
Step 5 · Incumbent

$$y^+ = \max\{\text{all measured values}\}$$

Step 6 · Improvement from measuring at x′

$$\text{improvement} = \max\left(f(x') - y^+,\ 0\right)$$





Step 7 · Expected improvement

$$\alpha(x') = \int_{y^+}^{\infty} \left(f(x') - y^+\right) p\left(f(x') \mid f(x)\right)\, df(x')$$

Jones, Schonlau & Welch (1998)

Georgia Tech

Step 8: evaluating the integral

Standard normal: density φ and its cumulative area Φ

$$\phi(s) = \frac{1}{\sqrt{2\pi}}\, e^{-s^2/2}, \qquad \Phi(s) = \int_{-\infty}^{s} \phi(t)\, dt$$

Split the improvement into two parts

$$f(x') - y^+ = \underbrace{\big(f(x') - m'(x')\big)}_{\text{part 1}} + \underbrace{\big(m'(x') - y^+\big)}_{\text{part 2: a constant}}$$

Part 2 × area above y^+ = probability of beating the incumbent

$$\int_{y^+}^{\infty} p\, df(x') = \Phi\left(\frac{m'(x') - y^+}{\sigma'(x')}\right)$$

Part 1: integrate using $dp/df = -(f - m')\, p / \sigma'^2$, so only the end point y^+ survives

$$\int_{y^+}^{\infty} \big(f(x') - m'(x')\big)\, p\, df(x') = \sigma'(x')^2\, p\big|_{f(x')=y^+} = \sigma'(x')\, \phi\left(\frac{m'(x') - y^+}{\sigma'(x')}\right)$$

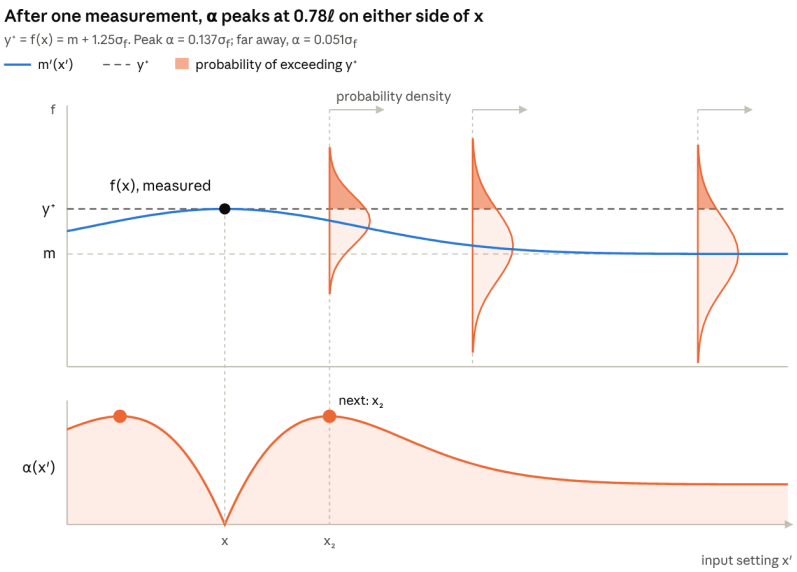
Takeaway Both integrals come out in φ and Φ, two standard functions evaluated to machine precision.

Georgia Tech

Step 9: expected improvement in closed form

$$\alpha(x') = \left(m'(x') - y^+\right)\Phi\left(\frac{m'(x') - y^+}{\sigma'(x')}\right) + \sigma'(x')\,\phi\left(\frac{m'(x') - y^+}{\sigma'(x')}\right)$$

large where m′ is above y^+ — large where σ′ is large



After one measurement

f(x) = m + 1.25σ_f, so y^+ = f(x).

At x: σ′ = 0, so α = 0. A measured setting is never chosen again.

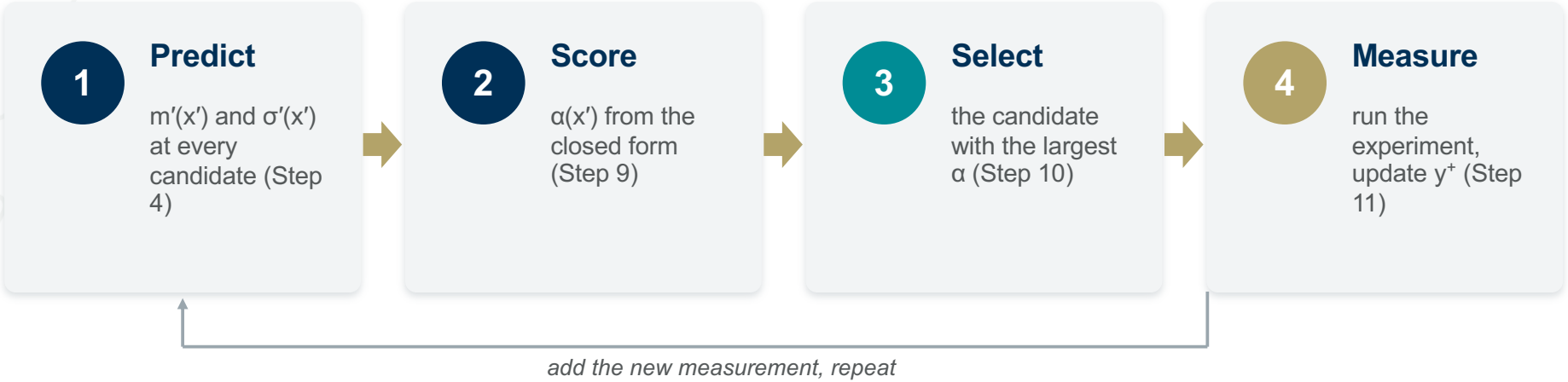
Peak: α = 0.137σ_f at 0.78ℓ on either side.

Far from x: α → 0.051σ_f.

Takeaway The right-hand peak becomes x_2: close enough to share the high value, far enough to be uncertain.

Georgia Tech

Steps 10–11: the loop



$$x_{\text{next}} = \arg\max_{x'} \alpha(x')$$

$$y^{+} \leftarrow \max\left(y^{+},\ f(x_{\text{next}})\right)$$

Steps 1–3 are arithmetic on a grid of candidates. Only step 4 costs a wafer.

Takeaway When the budget is spent, report the measured setting that produced y^{+}.

Georgia Tech

E

WORKED EXAMPLE

Seven measurements on a test function

Georgia Tech

Several measurements: one weight per measurement

Posterior mean

$$m'(x') = m + \sum_{i=1}^{n} w_i \left(f(x_i) - m \right)$$

Weights: least squares gives n linear equations

$$\sum_{j=1}^{n} k(x_i, x_j)\, w_j = k(x', x_i), \quad i = 1, \ldots, n \qquad \Longleftrightarrow \qquad K\, w = k(x')$$

Posterior variance

$$\sigma'(x')^2 = \sigma_f^2 - \sum_{i=1}^{n} w_i\, k(x', x_i)$$

Two measurements

$$w_1 = \frac{\sigma_f^2\, k(x', x_1) - k(x_1, x_2)\, k(x', x_2)}{\sigma_f^4 - k(x_1, x_2)^2}$$

Far apart: $k(x_1, x_2) = 0$, and each weight takes its single-measurement value.

Close together: $k(x_1, x_2) \to \sigma_f^2$, and each weight halves. Nearby points share the weight.

Takeaway Steps 5–11 are unchanged: conditioning on f(x) becomes conditioning on all measurements.

Georgia Tech

K w = k(x′) written out as matrices

$$\underbrace{\begin{bmatrix} k(x_1,x_1) & k(x_1,x_2) & \cdots & k(x_1,x_n) \\ k(x_2,x_1) & k(x_2,x_2) & \cdots & k(x_2,x_n) \\ \vdots & \vdots & \ddots & \vdots \\ k(x_n,x_1) & k(x_n,x_2) & \cdots & k(x_n,x_n) \end{bmatrix}}_{K\ (n\times n)} \underbrace{\begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}}_{w} = \underbrace{\begin{bmatrix} k(x',x_1) \\ k(x',x_2) \\ \vdots \\ k(x',x_n) \end{bmatrix}}_{k(x')}$$

Each row is one of the n equations

$$\text{row } i: \quad k(x_i,x_1)\,w_1 + k(x_i,x_2)\,w_2 + \cdots + k(x_i,x_n)\,w_n = k(x',x_i)$$

Each entry, squared-exponential kernel

$$K_{ij} = k(x_i,x_j) = \sigma_f^2\, e^{-(x_i-x_j)^2/2\ell^2}, \qquad k_i(x') = k(x',x_i) = \sigma_f^2\, e^{-(x'-x_i)^2/2\ell^2}$$

Reading K

Diagonal: $k(x_i, x_i) = \sigma_f^2$, each setting with itself.

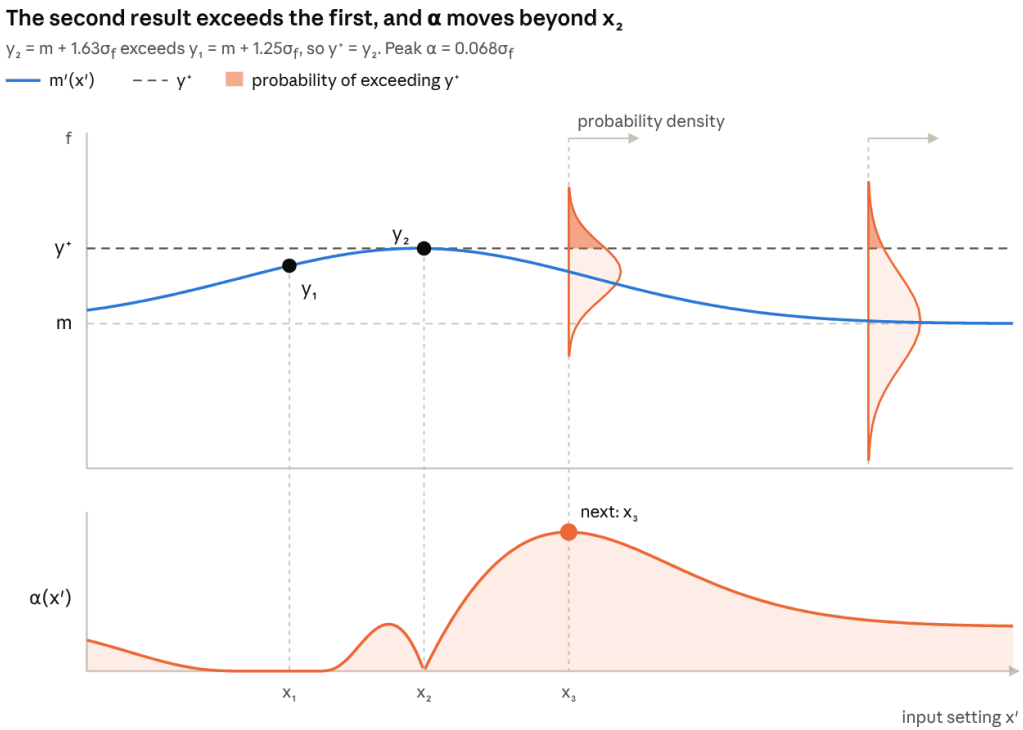
Symmetric: $k(x_i, x_j) = k(x_j, x_i)$.

Off-diagonal: the overlap between two measured settings; near 0 when they are far apart.

k(x′): the overlap of the candidate with each measured setting.

Takeaway K depends only on the measured settings; k(x′) changes as the candidate x′ moves.

Second measurement: a higher result



$x_2 = x_1 + 0.78\ell$

$y_2 = m + 1.63\sigma_f$, above $y_1 = m + 1.25\sigma_f$, so $y^+ = y_2$.

Between x_1 and x_2: narrow, below y_2; $\alpha < 0.023\sigma_f$.

0.83ℓ beyond x_2: still reaches above y_2; $\alpha = 0.068\sigma_f$, the maximum. This becomes x_3.

Right end: $\alpha = 0.022\sigma_f$.

Takeaway The search steps past the new best point, where the response may still be rising.

Georgia Tech

After y_2: the weights at x_3, worked by hand

Two measurements: K w = k(x′) written out

$$\begin{bmatrix} \sigma_f^2 & k(x_1,x_2) \\ k(x_1,x_2) & \sigma_f^2 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} k(x',x_1) \\ k(x',x_2) \end{bmatrix}$$

Squared-exponential kernel: divide every entry by σ_f^2

$$\rho_1 = e^{-(x'-x_1)^2/2\ell^2}, \quad \rho_2 = e^{-(x'-x_2)^2/2\ell^2}, \quad \rho_{12} = e^{-(x_1-x_2)^2/2\ell^2}$$

$$w_1 = \frac{\rho_1 - \rho_{12}\,\rho_2}{1-\rho_{12}^2}, \qquad w_2 = \frac{\rho_2 - \rho_{12}\,\rho_1}{1-\rho_{12}^2}, \qquad \sigma'(x')^2 = \sigma_f^2\left[1 - (w_1\rho_1 + w_2\rho_2)\right]$$

At $x_3 = x_2 + 0.83\ell$: $\rho_1 = 0.27$, $\rho_2 = 0.71$, $\rho_{12} = 0.74$

$$w_1 = \frac{0.27 - 0.74 \times 0.71}{1 - 0.74^2} = -0.55, \qquad w_2 = \frac{0.71 - 0.74 \times 0.27}{1 - 0.74^2} = 1.12$$

$m'(x_3) = m + (-0.55)(1.25\sigma_f) + (1.12)(1.63\sigma_f) = m + 1.12\sigma_f$ $\quad$ $\sigma'(x_3) = 0.60\sigma_f$

Takeaway w_1 is negative: x_1 and x_2 overlap ($\rho_{12} = 0.74$), so the solve leans on y_2 and subtracts part of y_1.

Georgia Tech

After y_2: the same solve as a matrix

Matrix element

$$K_{ij} = k(x_i, x_j) = \sigma_f^2\, e^{-(x_i - x_j)^2/2\ell^2}, \qquad k_i(x') = k(x', x_i) = \sigma_f^2\, e^{-(x' - x_i)^2/2\ell^2}$$

Settings, in ℓ from x_1: $x_1 = 0.00$, $x_2 = 0.78$ **candidate $x' = x_3 = 1.61$**

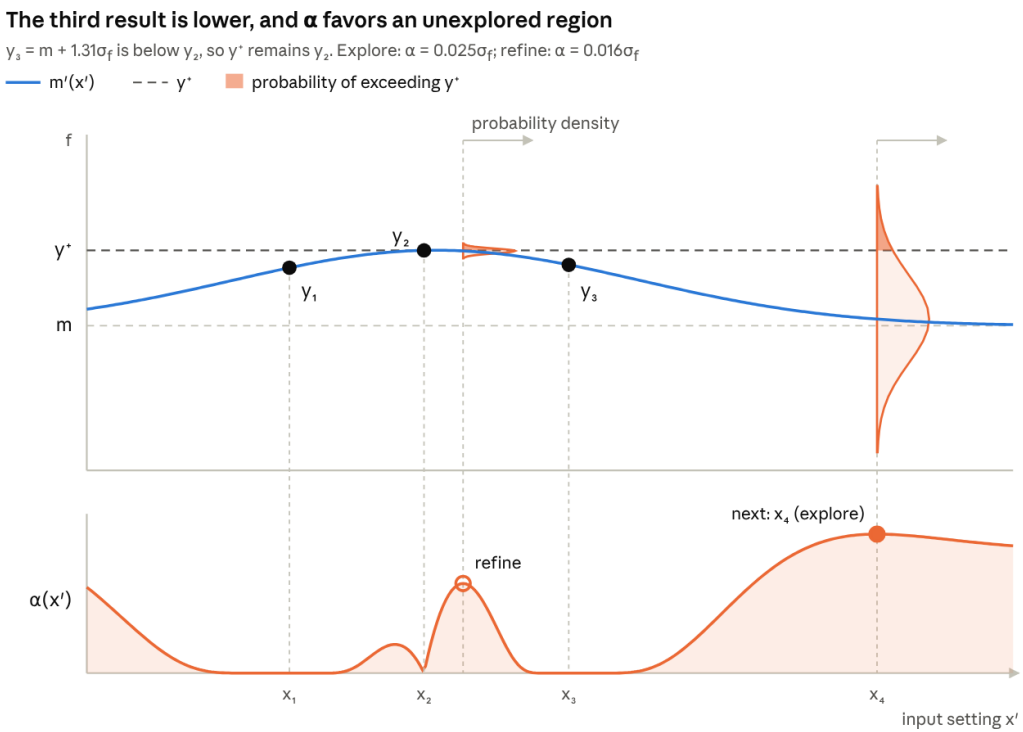
$$\underbrace{\begin{bmatrix} 1 & 0.74 \\ 0.74 & 1 \end{bmatrix}}_{K/\sigma_f^2} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0.27 \\ 0.71 \end{bmatrix}}_{k(x')/\sigma_f^2} \quad \Rightarrow \quad w = \begin{bmatrix} -0.55 \\ 1.12 \end{bmatrix} \qquad \frac{y - m}{\sigma_f} = \begin{bmatrix} 1.25 \\ 1.63 \end{bmatrix}$$

$$m'(x') = m + \sum_i w_i\,(y_i - m) = m + 1.12\,\sigma_f, \qquad \sigma'(x') = \sigma_f \sqrt{1 - \sum_i w_i\, k_i(x')/\sigma_f^2} = 0.60\,\sigma_f$$

Takeaway The same numbers in matrix form. Each new measurement adds one row and one column to K.

Georgia Tech

Third measurement: a lower result



$x_3 = x_2 + 0.83\ell$

$y_3 = m + 1.31\sigma_f$, below y_2: the incumbent stays y_2.

m′(x′) now passes through all three points.

Narrow candidate, 0.23ℓ beyond x_2: $\alpha = 0.016\sigma_f$.

Wide candidate, 1.78ℓ beyond x_3: $\alpha = 0.025\sigma_f$.

Takeaway Two candidates compete: a narrow one beside the best result and a wide one far from all data.

Georgia Tech

After y_3: the weights at x_4

Matrix element

$$K_{ij} = k(x_i, x_j) = \sigma_f^2\, e^{-(x_i - x_j)^2/2\ell^2}, \qquad k_i(x') = k(x', x_i) = \sigma_f^2\, e^{-(x' - x_i)^2/2\ell^2}$$

Settings, in ℓ from x_1: x_1 = 0.00, x_2 = 0.78, x_3 = 1.61 **candidate $x' = x_4$ = 3.38**

$$\underbrace{\begin{bmatrix} 1 & 0.74 & 0.27 \\ 0.74 & 1 & 0.71 \\ 0.27 & 0.71 & 1 \end{bmatrix}}_{K/\sigma_f^2} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 0.00 \\ 0.03 \\ 0.21 \end{bmatrix}}_{k(x')/\sigma_f^2} \quad \Rightarrow \quad w = \begin{bmatrix} 0.21 \\ -0.46 \\ 0.47 \end{bmatrix} \qquad \frac{y - m}{\sigma_f} = \begin{bmatrix} 1.25 \\ 1.63 \\ 1.31 \end{bmatrix}$$

$$m'(x') = m + \sum_i w_i\,(y_i - m) = m + 0.14\,\sigma_f, \qquad \sigma'(x') = \sigma_f \sqrt{1 - \sum_i w_i\, k_i(x')/\sigma_f^2} = 0.96\,\sigma_f$$

Takeaway x_4 is far from every measurement: the weights sum to 0.23, and m′ stays near m.

Georgia Tech

Exploration versus exploitation

Candidate after three measurements	m′(x′)	σ′(x′)	α(x′)
Exploitation: 0.23ℓ beyond x_2	m + 1.61σ_f	0.05σ_f	0.016σ_f
Exploration: 1.78ℓ beyond x_3	m + 0.14σ_f	0.96σ_f	0.025σ_f

$$\frac{\partial \alpha}{\partial m'(x')} = \Phi\left(\frac{m'(x') - y^+}{\sigma'(x')}\right), \qquad \frac{\partial \alpha}{\partial \sigma'(x')} = \phi\left(\frac{m'(x') - y^+}{\sigma'(x')}\right)$$

Both derivatives are positive. At $m' = y^+$: $\Phi = 0.50$, $\varphi = 0.40$. At $m' = y^+ - 1.5\sigma'$: $\Phi = 0.067$, $\varphi = 0.13$, so spread counts twice as much as mean.

1

Exploitation

Measure where m′ is high, beside the best result. Alone, it can settle on a local maximum.

2

Exploration

Measure where σ′ is large, far from the data. Alone, it spends runs where f is already predicted low.

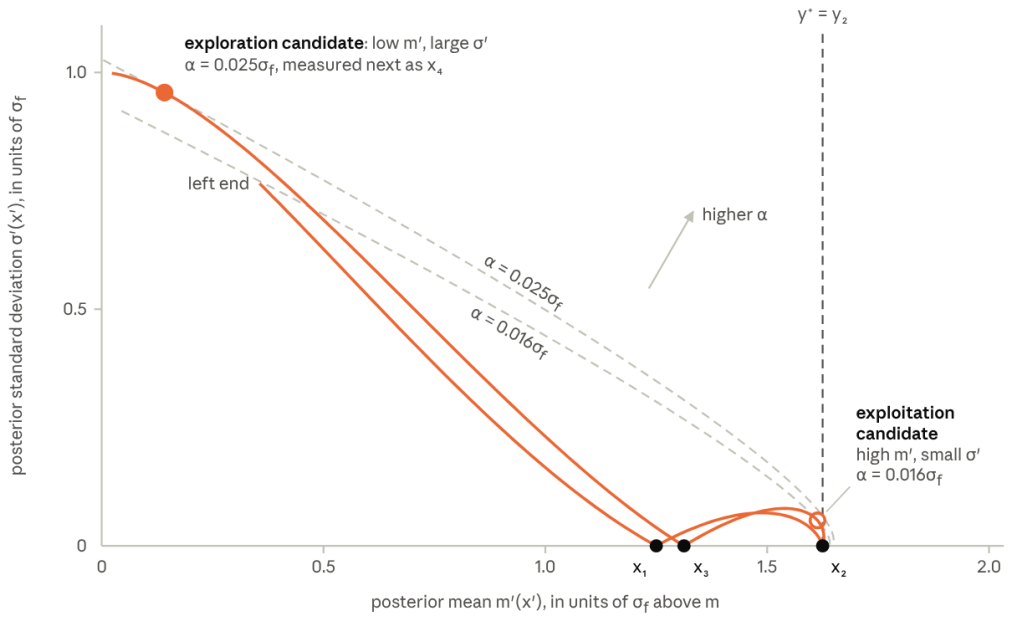
Takeaway Expected improvement needs no tuning knob: the ratio of Φ to φ sets the exchange.

Georgia Tech

Every candidate in the plane of m′ and σ′

α rises with both the posterior mean and the posterior standard deviation

Candidate settings x′ after the third measurement, placed by m′(x′) and σ′(x′); dashed lines of constant α



Reading the figure

Dashed lines: constant α. Along a line, a lower mean is offset by a wider spread.

The curve of candidates touches the highest line only at the exploration candidate, which becomes x_4.

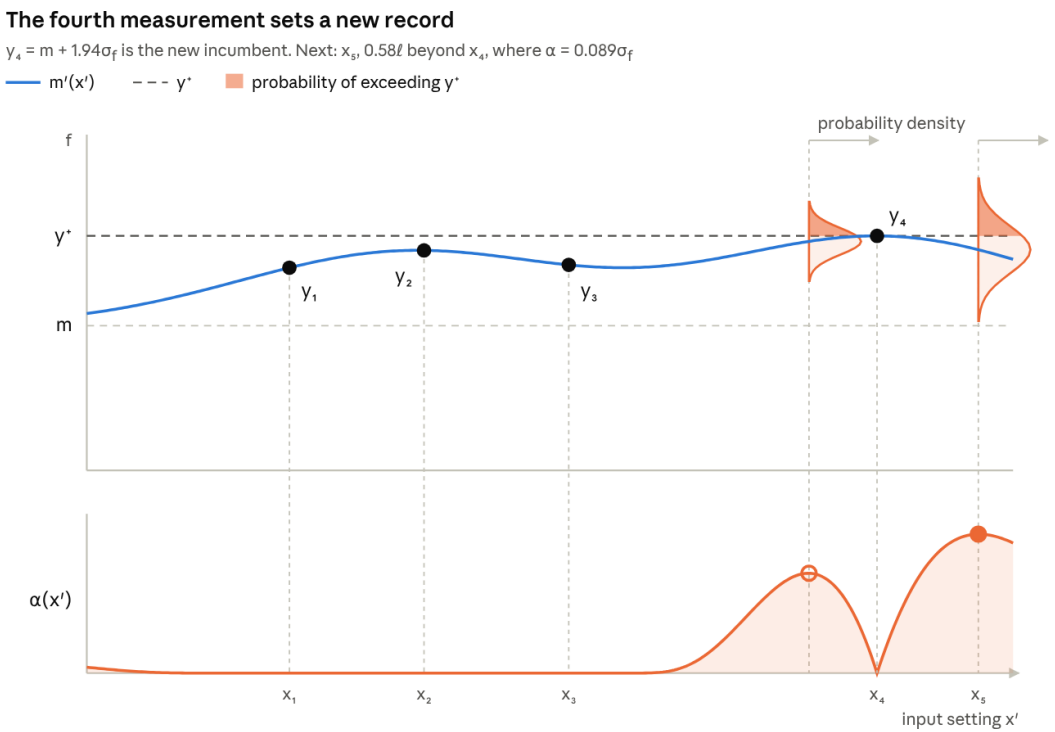
Upper confidence bound (Srinivas et al., 2010) sets the balance by hand:

UCB: choose the largest $m'(x') + \beta\,\sigma'(x')$

Takeaway The chosen point is where the curve of candidates touches the highest line of constant α.

Georgia Tech

Fourth measurement: exploration finds a better region



$x_4 = x_3 + 1.78\ell$

$y_4 = m + 1.94\sigma_f$, above y_2: the new incumbent.

0.39ℓ before x_4: $m' = m + 1.82\sigma_f$, $\sigma' = 0.28\sigma_f$, $\alpha = 0.064\sigma_f$.

0.58ℓ beyond x_4: $m' = m + 1.64\sigma_f$, $\sigma' = 0.51\sigma_f$, $\alpha = 0.089\sigma_f$. This becomes x_5.

Takeaway A setting chosen for the width of its distribution produced the new best result.

Georgia Tech

After y_4: the weights at x_5

Matrix element

$$K_{ij} = k(x_i, x_j) = \sigma_f^2\, e^{-(x_i - x_j)^2/2\ell^2}, \qquad k_i(x') = k(x', x_i) = \sigma_f^2\, e^{-(x' - x_i)^2/2\ell^2}$$

Settings, in ℓ from x_1: $x_1 = 0.00$, $x_2 = 0.78$, $x_3 = 1.61$, $x_4 = 3.38$ **candidate $x' = x_5 = 3.97$**

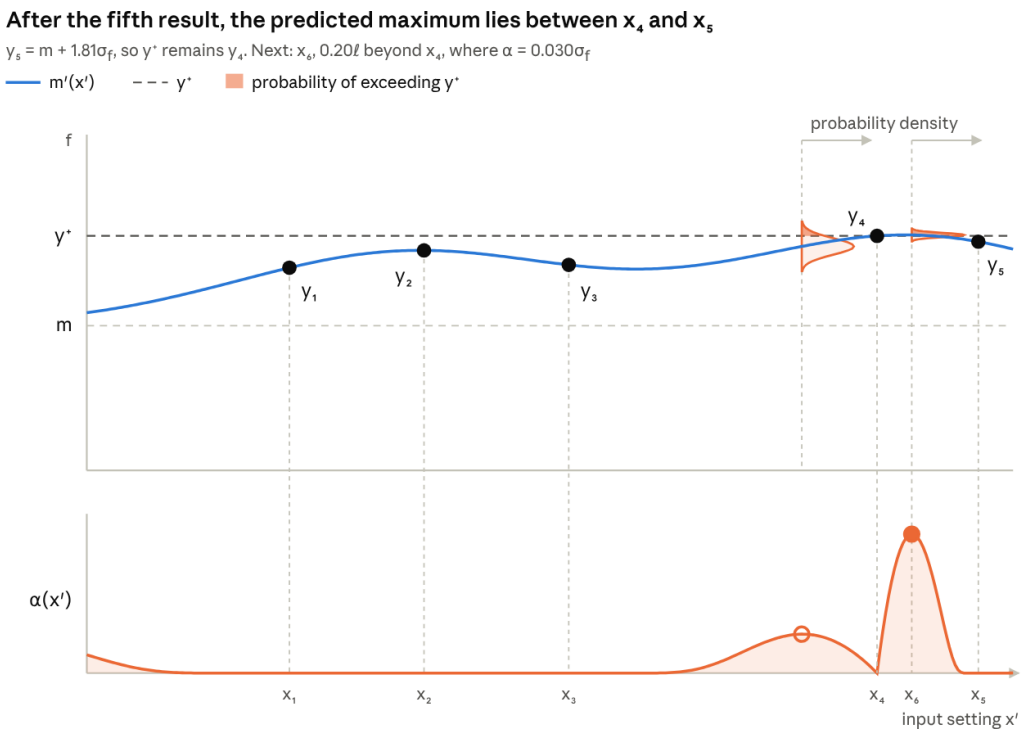
$$\underbrace{\begin{bmatrix} 1 & 0.74 & 0.27 & 0.00 \\ 0.74 & 1 & 0.71 & 0.03 \\ 0.27 & 0.71 & 1 & 0.21 \\ 0.00 & 0.03 & 0.21 & 1 \end{bmatrix}}_{K/\sigma_f^2} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix} = \underbrace{\begin{bmatrix} 0.00 \\ 0.01 \\ 0.06 \\ 0.84 \end{bmatrix}}_{k(x')/\sigma_f^2} \quad \Rightarrow \quad w = \begin{bmatrix} -0.12 \\ 0.25 \\ -0.27 \\ 0.89 \end{bmatrix} \qquad \frac{y - m}{\sigma_f} = \begin{bmatrix} 1.25 \\ 1.63 \\ 1.31 \\ 1.94 \end{bmatrix}$$

$$m'(x') = m + \sum_i w_i\,(y_i - m) = m + 1.64\,\sigma_f, \qquad \sigma'(x') = \sigma_f \sqrt{1 - \sum_i w_i\, k_i(x')/\sigma_f^2} = 0.51\,\sigma_f$$

Takeaway x_5 is 0.58ℓ from x_4, so $w_4 = 0.89$ dominates; the other weights are small.

Georgia Tech

Fifth measurement: the mean curves over



$x_5 = x_4 + 0.58\ell$

$y_5 = m + 1.81\sigma_f$, below y_4.

m′ peaks 0.16ℓ beyond x_4 at $m + 1.96\sigma_f$, just above y^+.

α peaks 0.20ℓ beyond x_4: $0.030\sigma_f$, with $\sigma' = 0.05\sigma_f$ and two-thirds of the distribution above y^+. This becomes x_6.

0.43ℓ before x_4: α has fallen to $0.008\sigma_f$.

Takeaway Two high results bracket a peak, and the search measures between them.

Georgia Tech

After y_5: the weights at x_6

Matrix element

$$K_{ij} = k(x_i, x_j) = \sigma_f^2\, e^{-(x_i - x_j)^2/2\ell^2}, \qquad k_i(x') = k(x', x_i) = \sigma_f^2\, e^{-(x' - x_i)^2/2\ell^2}$$

Settings, in ℓ from x_1: x_1 = 0.00, x_2 = 0.78, x_3 = 1.61, x_4 = 3.38, x_5 = 3.97 **candidate $x' = x_6$ = 3.58**

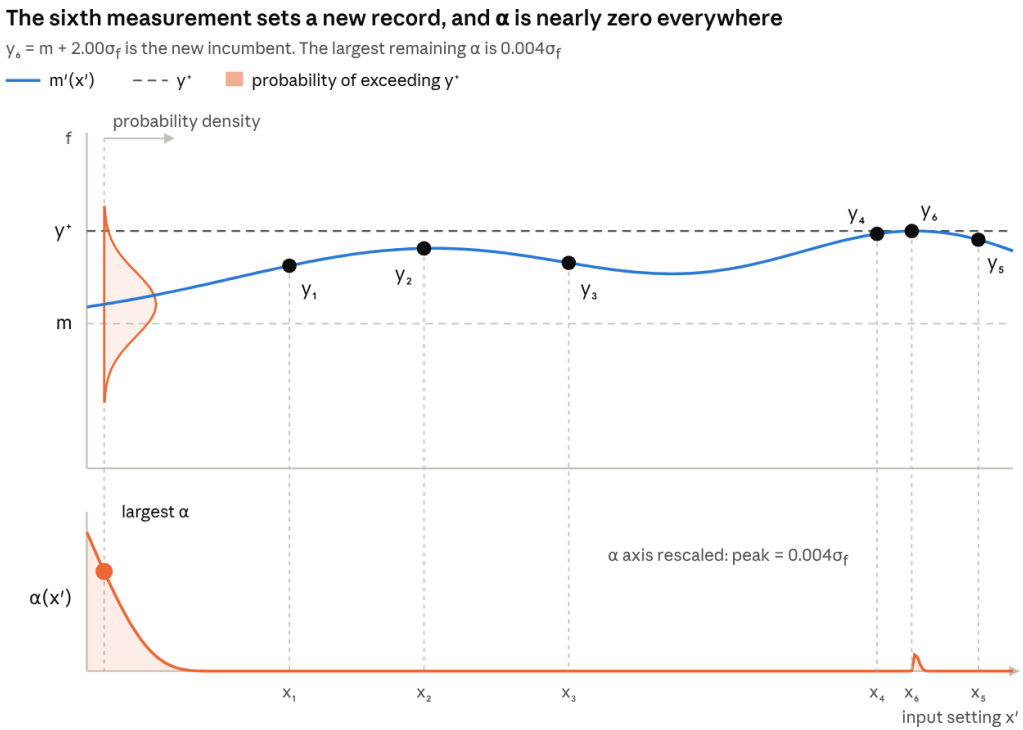
$$\underbrace{\begin{bmatrix} 1 & 0.74 & 0.27 & 0.00 & 0.00 \\ 0.74 & 1 & 0.71 & 0.03 & 0.01 \\ 0.27 & 0.71 & 1 & 0.21 & 0.06 \\ 0.00 & 0.03 & 0.21 & 1 & 0.84 \\ 0.00 & 0.01 & 0.06 & 0.84 & 1 \end{bmatrix}}_{K/\sigma_f^2} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \end{bmatrix} = \underbrace{\begin{bmatrix} 0.00 \\ 0.02 \\ 0.14 \\ 0.98 \\ 0.93 \end{bmatrix}}_{k(x')/\sigma_f^2} \quad \Rightarrow \quad w = \begin{bmatrix} -0.02 \\ 0.04 \\ -0.05 \\ 0.70 \\ 0.34 \end{bmatrix} \qquad \frac{y - m}{\sigma_f} = \begin{bmatrix} 1.25 \\ 1.63 \\ 1.31 \\ 1.94 \\ 1.81 \end{bmatrix}$$

$$m'(x') = m + \sum_i w_i\,(y_i - m) = m + 1.96\,\sigma_f, \qquad \sigma'(x') = \sigma_f \sqrt{1 - \sum_i w_i\, k_i(x')/\sigma_f^2} = 0.05\,\sigma_f$$

Takeaway x_6 lies between x_4 and x_5: w_4 = 0.70 and w_5 = 0.34 carry the estimate.

Georgia Tech

Sixth measurement: a new incumbent



$x_6 = x_4 + 0.20\ell$

$y_6 = m + 2.00\sigma_f$, above y_4: the new incumbent.

Between x_4, x_6 and x_5: σ' below $0.01\sigma_f$.

Largest α left: $0.004\sigma_f$, at the left end, where $m' = m + 0.36\sigma_f$ and $\sigma' = 0.76\sigma_f$.

Takeaway Every remaining distribution is either narrow or centered well below the best result.

Georgia Tech

After y_6: the weights at x_7, the left end

Matrix element

$$K_{ij} = k(x_i, x_j) = \sigma_f^2\, e^{-(x_i - x_j)^2/2\ell^2}, \qquad k_i(x') = k(x', x_i) = \sigma_f^2\, e^{-(x' - x_i)^2/2\ell^2}$$

Settings, in ℓ from x_1: $x_1 = 0.00$, $x_2 = 0.78$, $x_3 = 1.61$, $x_4 = 3.38$, $x_5 = 3.97$, $x_6 = 3.58$ **candidate $x' = x_7 = -1.17$**

$$\underbrace{\begin{bmatrix} 1 & 0.74 & 0.27 & 0.00 & 0.00 & 0.00 \\ 0.74 & 1 & 0.71 & 0.03 & 0.01 & 0.02 \\ 0.27 & 0.71 & 1 & 0.21 & 0.06 & 0.14 \\ 0.00 & 0.03 & 0.21 & 1 & 0.84 & 0.98 \\ 0.00 & 0.01 & 0.06 & 0.84 & 1 & 0.93 \\ 0.00 & 0.02 & 0.14 & 0.98 & 0.93 & 1 \end{bmatrix}}_{K/\sigma_f^2} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \\ w_6 \end{bmatrix} = \underbrace{\begin{bmatrix} 0.51 \\ 0.15 \\ 0.02 \\ 0.00 \\ 0.00 \\ 0.00 \end{bmatrix}}_{k(x')/\sigma_f^2} \quad \Rightarrow \quad w = \begin{bmatrix} 1.13 \\ -1.03 \\ 0.51 \\ -1.52 \\ -0.56 \\ 1.96 \end{bmatrix} \qquad \frac{y - m}{\sigma_f} = \begin{bmatrix} 1.25 \\ 1.63 \\ 1.31 \\ 1.94 \\ 1.81 \\ 2.00 \end{bmatrix}$$

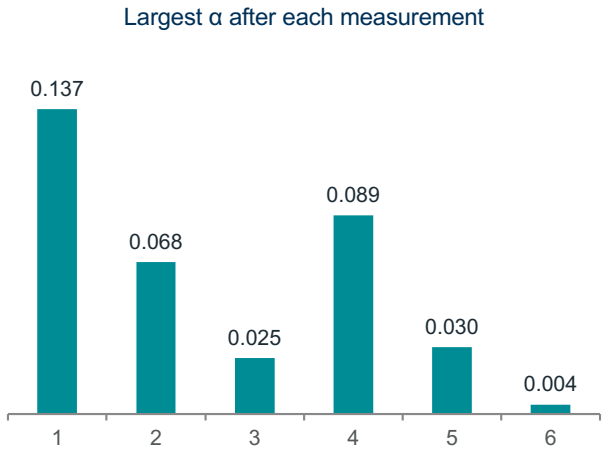
$$m'(x') = m + \sum_i w_i\,(y_i - m) = m + 0.36\,\sigma_f, \qquad \sigma'(x') = \sigma_f \sqrt{1 - \sum_i w_i\, k_i(x')/\sigma_f^2} = 0.76\,\sigma_f$$

Takeaway x_4, x_5 and x_6 get large weights of opposite sign; together they add only $-0.05\sigma_f$ to m′.

Georgia Tech

The record so far, and when to stop

n	Setting	Result	Incumbent	Max α after
1	x_1	$m + 1.25\sigma_f$	y_1	$0.137\sigma_f$
2	$x_2 = x_1 + 0.78\ell$	$m + 1.63\sigma_f$	y_2	$0.068\sigma_f$
3	$x_3 = x_2 + 0.83\ell$	$m + 1.31\sigma_f$	y_2	$0.025\sigma_f$
4	$x_4 = x_3 + 1.78\ell$	$m + 1.94\sigma_f$	y_4	$0.089\sigma_f$
5	$x_5 = x_4 + 0.58\ell$	$m + 1.81\sigma_f$	y_4	$0.030\sigma_f$
6	$x_6 = x_4 + 0.20\ell$	$m + 2.00\sigma_f$	y_6	$0.004\sigma_f$



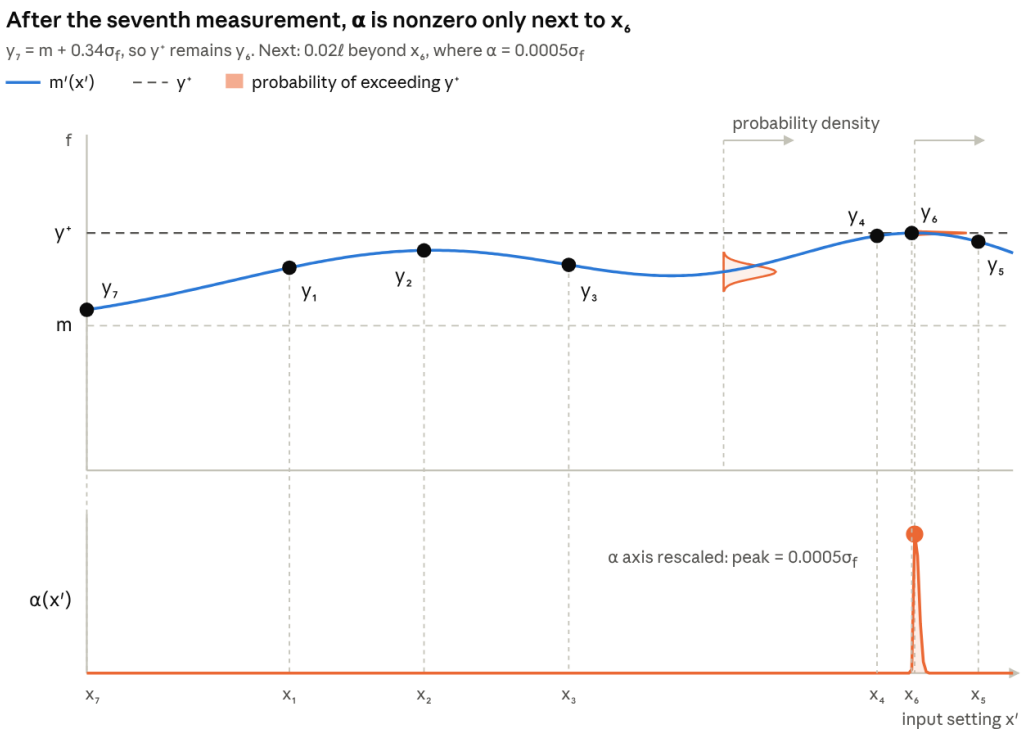
Stopping rule

Stop when the largest α is worth less than one more measurement. Here α fell to $0.004\sigma_f$, 3% of its first value, so the search reports x_6 with $y^+ = m + 2.00\sigma_f$. The hidden test function peaks at $m + 2.01\sigma_f$, 0.03ℓ from x_6.

Takeaway Six measurements located the maximum to within $0.01\sigma_f$.

Georgia Tech

Seventh measurement and termination



Continuing past the stopping rule

x_7: the left end, 1.17ℓ from x_1, the widest distribution left.

$y_7 = m + 0.34\sigma_f$; the prediction was $m + 0.36\sigma_f$.

Largest α left: $0.0005\sigma_f$, immediately beside x_6.

Readout resolution $0.03\sigma_f$, 63 times larger. No measurement can resolve the gain.

Takeaway The seventh run confirmed the prediction at the left end and changed nothing else.

Georgia Tech

After y_7: the weights beside x_6

Matrix element

$$K_{ij} = k(x_i, x_j) = \sigma_f^2\, e^{-(x_i - x_j)^2/2\ell^2}, \qquad k_i(x') = k(x', x_i) = \sigma_f^2\, e^{-(x' - x_i)^2/2\ell^2}$$

Settings, in ℓ from x_1: $x_1 = 0.00$, $x_2 = 0.78$, $x_3 = 1.61$, $x_4 = 3.38$, $x_5 = 3.97$, $x_6 = 3.58$, $x_7 = -1.17$ **candidate $x' = x_8 = 3.61$**

$$\underbrace{\begin{bmatrix} 1 & 0.74 & 0.27 & 0.00 & 0.00 & 0.00 & 0.51 \\ 0.74 & 1 & 0.71 & 0.03 & 0.01 & 0.02 & 0.15 \\ 0.27 & 0.71 & 1 & 0.21 & 0.06 & 0.14 & 0.02 \\ 0.00 & 0.03 & 0.21 & 1 & 0.84 & 0.98 & 0.00 \\ 0.00 & 0.01 & 0.06 & 0.84 & 1 & 0.93 & 0.00 \\ 0.00 & 0.02 & 0.14 & 0.98 & 0.93 & 1 & 0.00 \\ 0.51 & 0.15 & 0.02 & 0.00 & 0.00 & 0.00 & 1 \end{bmatrix}}_{K/\sigma_f^2} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \\ w_6 \\ w_7 \end{bmatrix} = \underbrace{\begin{bmatrix} 0.00 \\ 0.02 \\ 0.14 \\ 0.98 \\ 0.94 \\ 1.00 \\ 0.00 \end{bmatrix}}_{k(x')/\sigma_f^2} \quad \Rightarrow \quad w = \begin{bmatrix} 0.00 \\ 0.00 \\ 0.00 \\ -0.08 \\ 0.02 \\ 1.06 \\ 0.00 \end{bmatrix} \qquad \frac{y - m}{\sigma_f} = \begin{bmatrix} 1.25 \\ 1.63 \\ 1.31 \\ 1.94 \\ 1.81 \\ 2.00 \\ 0.34 \end{bmatrix}$$

$$m'(x') = m + \sum_i w_i\,(y_i - m) = m + 2.00\,\sigma_f, \qquad \sigma'(x') = \sigma_f \sqrt{1 - \sum_i w_i\, k_i(x')/\sigma_f^2} = 0.001\,\sigma_f$$

Takeaway Next to x_6, $w_6 = 1.06$ carries the estimate: $m' = m + 2.00\sigma_f$, and σ' is $0.001\sigma_f$.

Georgia Tech

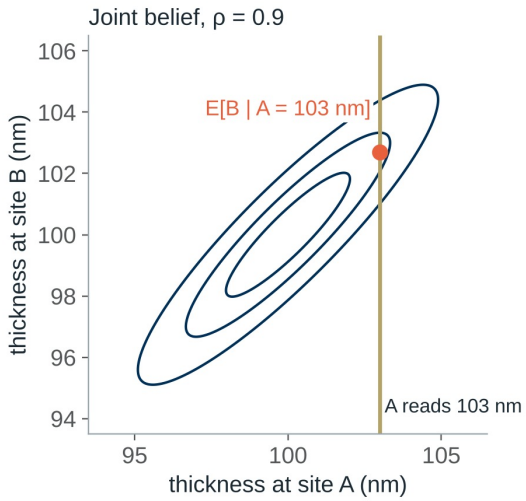
F

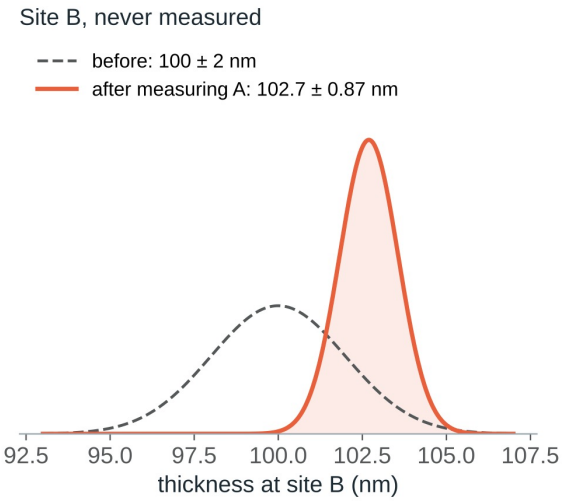
BACK TO THE FAB

Wafer thickness from a few sites

Georgia Tech

Two wafer sites: one measurement predicts the other





Step 4 with real units

Prior: m = 100 nm, σ_f = 2 nm at every site.

Kernel: sites 18 mm apart with ℓ = 40 mm: ρ = 0.9, k(A, B) = 3.6 nm².

Measure A: 103 nm.

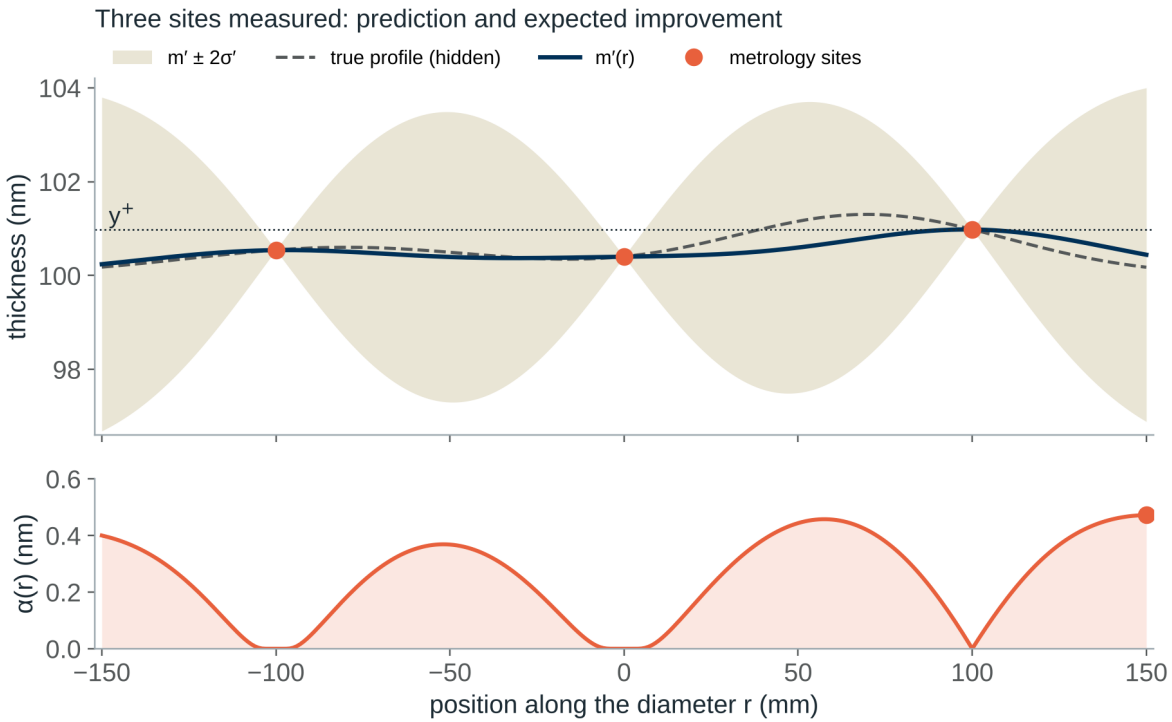
Weight: $w = k/\sigma_f^2 = 0.9$.

Site B: $m' = 102.7$ nm, $\sigma' = 0.87$ nm.

Takeaway Site B was never measured. Its estimate moved 2.7 nm and its error bar fell by more than half.

Georgia Tech

Where is the film thickest? Three sites to start



One diameter, 300 mm wafer

Sites: r = −100, 0, +100 mm; best 100.98 nm at +100 mm.

Same prior: 100 ± 2 nm, ℓ = 40 mm; readout to 0.01 nm.

Gold band: m′ ± 2σ′. It closes at each site and opens to ±3.6 nm at the edge.

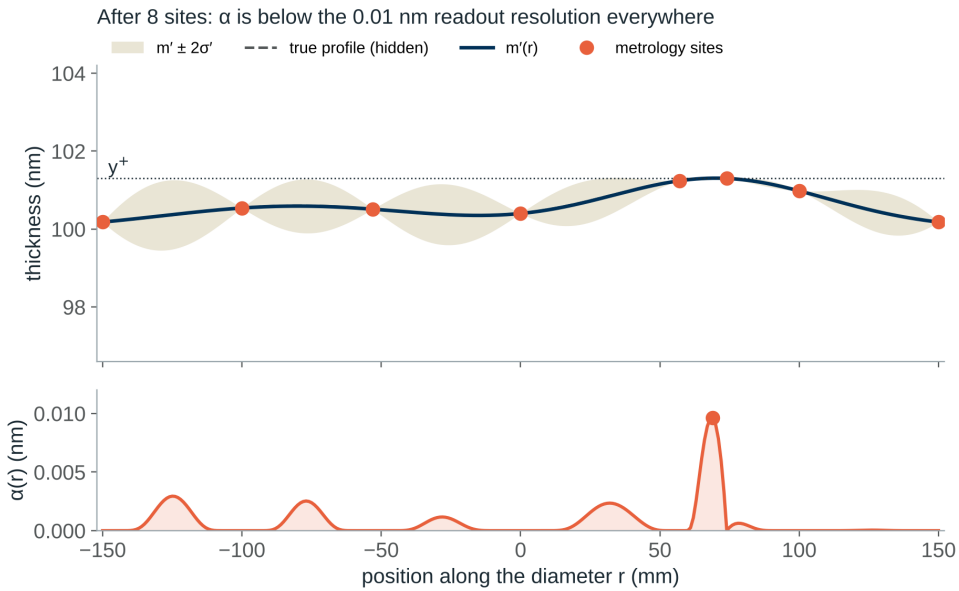
Largest α: 0.47 nm at the edge, r = +150 mm, where σ′ = 1.78 nm.

Takeaway The first pick is pure exploration: the wafer edge, farthest from every site.

Georgia Tech

The search, site by site, and where it stops

Site	r (mm)	Thickness	y^+ after	Max α after
1	-100	100.54 nm	100.54 nm	–
2	0	100.40 nm	100.54 nm	–
3	+100	100.98 nm	100.98 nm	0.47 nm
4	+150	100.18 nm	100.98 nm	0.46 nm
5	+57	101.24 nm	101.24 nm	0.32 nm
6	-150	100.18 nm	101.24 nm	0.23 nm
7	-53	100.51 nm	101.24 nm	0.13 nm
8	+74	101.30 nm	101.30 nm	0.0096 nm



Stop: after site 8, the largest α is 0.0096 nm, below the 0.01 nm readout. Reported: 101.30 nm at r = +74 mm. True maximum: 101.30 nm at +70 mm.

Takeaway 5 extra sites, chosen by expected improvement, found the thickest point to the readout resolution.

Georgia Tech

G

PERSPECTIVE

Other methods, history, and when to use BO

Georgia Tech

Other ways to choose the wafers, on the same anneal problem

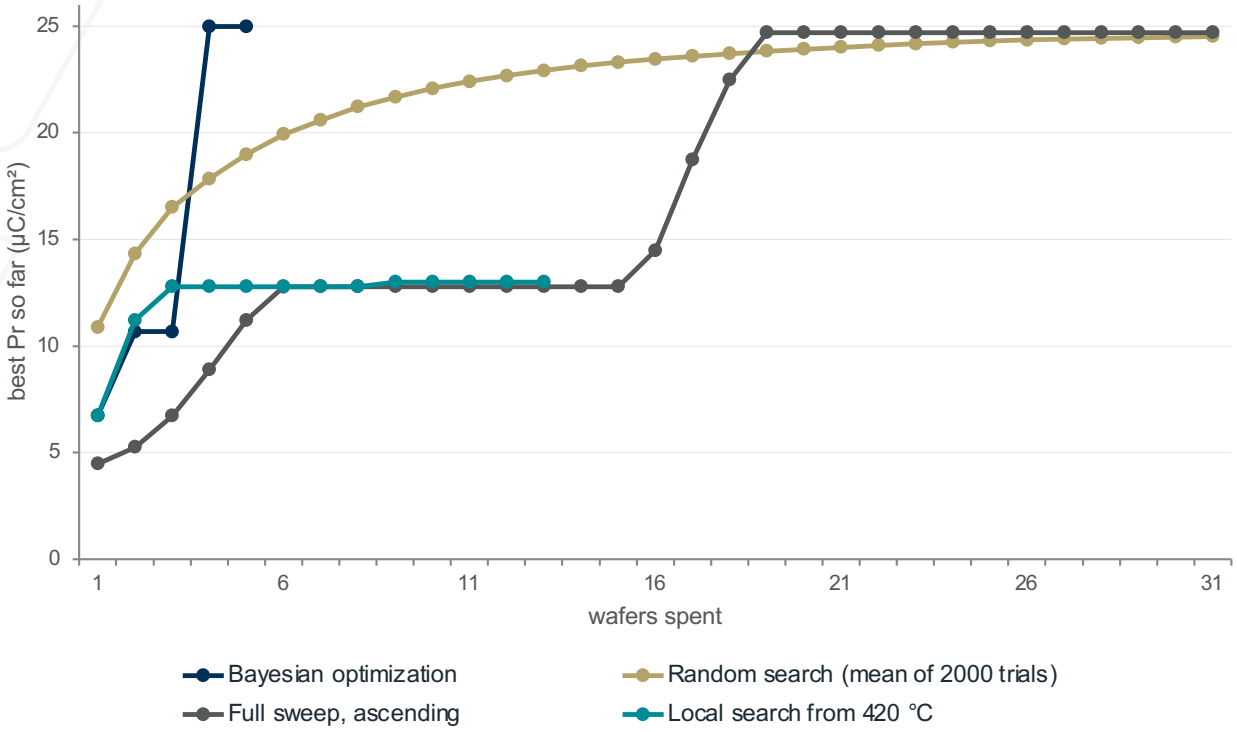
Method	How it chooses the wafers	Wafers	Best Pr (μC/cm²)	Within 1 μC/cm² of optimum
Full sweep, 10 °C steps	every temperature on a fixed grid	31	24.7 at 580 °C	from wafer 19, in ascending order
Coarse grid, 50 °C steps	a fixed grid of seven temperatures	7	22.5 at 600 °C	no
Random search	temperatures drawn uniformly at random	6	19.9, mean of 2000 trials	31% of trials
Design of experiments + quadratic	five levels, fit a parabola, confirm at its peak (552 °C)	6	15.4	no
Local search	climb from 420 °C in 20 °C steps, halve the step when stuck	13	13.0, the local peak at 455 °C	no
Bayesian optimization	the loop of this lecture, from 420, 540 and 680 °C	5	25.0 at 586 °C	from wafer 4

Optimum: 25.0 μC/cm² at 585 °C. BO kernel: σ_f = 7 μC/cm², ℓ = 42 °C, prior mean 7.2 μC/cm² (average of the three starting wafers); stopped when the largest α fell below 0.1 μC/cm².

Takeaway Bayesian optimization reached the optimum in 5 wafers. The sweep needed 19 in order and 31 to finish.

Georgia Tech

Best result against wafers spent



Reading the chart

BO: 25.0 µC/cm² on wafer 4; stops after wafer 5.

Random: mean 19.9 after 6 wafers, 24.5 after 31.

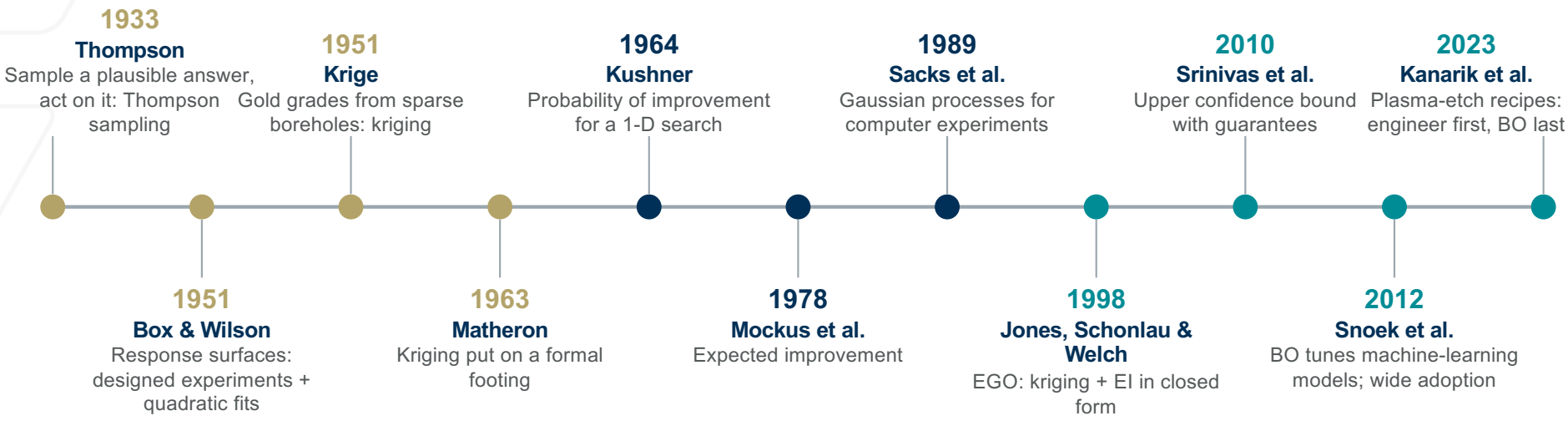
Sweep: flat until it reaches the peak region at wafer 19.

Local search: stuck at 13.0 µC/cm², the local peak.

Takeaway Each method spends wafers differently. Only BO uses every result to decide the next temperature.

Georgia Tech

A short history of Bayesian optimization



Takeaway The model is kriging from 1951; the decision rule is expected improvement from 1978. EGO (1998) joined them.

Georgia Tech

When to use Bayesian optimization, and when not to

Use it when

- **Each run is expensive**
  a wafer, hours of tool time, a long simulation
- **The budget is tens of runs**
  every result has to inform the next choice
- **A handful of knobs**
  standard BO is used below about 20 inputs (Frazier, 2018)
- **A smooth response**
  smooth over a correlation length the budget can resolve
- **No gradient**
  the process returns a value and nothing else

Look elsewhere when

- **Runs are cheap**
  a sweep or a population method costs less in total
- **Many knobs**
  separations stop being informative; the model needs many runs
- **Thousands of runs**
  solving $K w = k(x')$ grows as the cube of the number of runs
- **Cliffs and failed runs**
  a smooth kernel blurs window edges; a failed run returns no number
- **Features narrower than ℓ**
  the error bar covers only the variation the kernel allows

Takeaway The sweet spot is a few process knobs and a budget of tens of expensive runs.

Georgia Tech

Summary

1. **Correlation is the link:** a distribution at one setting says nothing about its neighbors; the kernel $k(x, x')$ does.

2. **Prediction is least squares:** m' moves from m toward each measurement by its weight; σ' shrinks within about ℓ of the data.

3. **Expected improvement is the average of a rectified gain:** $\alpha = (m' - y^{+})\Phi + \sigma'\varphi$, zero at every measured setting.

4. **Mean and spread both raise α:** $\partial\alpha/\partial m' = \Phi$ and $\partial\alpha/\partial\sigma' = \varphi$; their ratio balances exploitation and exploration.

5. **Stop on a number:** when the largest α is worth less than one measurement, report the best measured setting.

Georgia Tech

References

W. R. Thompson, "On the likelihood that one unknown probability exceeds another in view of the evidence of two samples," *Biometrika*, vol. 25, pp. 285–294, 1933.

G. E. P. Box and K. B. Wilson, "On the experimental attainment of optimum conditions," *J. Royal Statistical Society B*, vol. 13, pp. 1–45, 1951.

D. G. Krige, "A statistical approach to some basic mine valuation problems on the Witwatersrand," *J. Chem. Metall. Min. Soc. South Africa*, vol. 52, pp. 119–139, 1951.

G. Matheron, "Principles of geostatistics," *Economic Geology*, vol. 58, pp. 1246–1266, 1963.

H. J. Kushner, "A new method of locating the maximum point of an arbitrary multipeak curve in the presence of noise," *J. Basic Engineering*, vol. 86, pp. 97–106, 1964.

J. Mockus, V. Tiesis and A. Žilinskas, "The application of Bayesian methods for seeking the extremum," *Towards Global Optimization 2*, pp. 117–129, 1978.

J. Sacks, W. J. Welch, T. J. Mitchell and H. P. Wynn, "Design and analysis of computer experiments," *Statistical Science*, vol. 4, pp. 409–423, 1989.

D. R. Jones, M. Schonlau and W. J. Welch, "Efficient global optimization of expensive black-box functions," *J. Global Optimization*, vol. 13, pp. 455–492, 1998.

C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning*. MIT Press, 2006.

N. Srinivas, A. Krause, S. M. Kakade and M. Seeger, "Gaussian process optimization in the bandit setting: No regret and experimental design," *Proc. ICML*, pp. 1015–1022, 2010.

J. Snoek, H. Larochelle and R. P. Adams, "Practical Bayesian optimization of machine learning algorithms," *Advances in Neural Information Processing Systems 25*, 2012.

P. I. Frazier, "A tutorial on Bayesian optimization," *arXiv:1807.02811*, 2018.

K. J. Kanarik et al., "Human–machine collaboration for improving semiconductor process development," *Nature*, vol. 616, pp. 707–711, 2023.

Georgia Tech